

Project Proposal



Online Attendance & Payroll System for MIA Enterprises through RFID (OA&PS RFID)

Supervised by

Faculty of Applied Sciences

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Faculty of Applied Sciences,
International Islamic University, Islamabad.



Read In the name of your Lord Who created you...

(IQRA-1, Holy Quran)

INTRODUCTION

1. **SMART LABELS**
2. **SCOPE OF PROJECT**
3. **RFID TECHNOLOGY (Main Focus)**
4. **RFID VARIOUS APPLICATIONS**
5. **QUESTIONS & ANSWERS SESSION**



- **What are smart labels?**

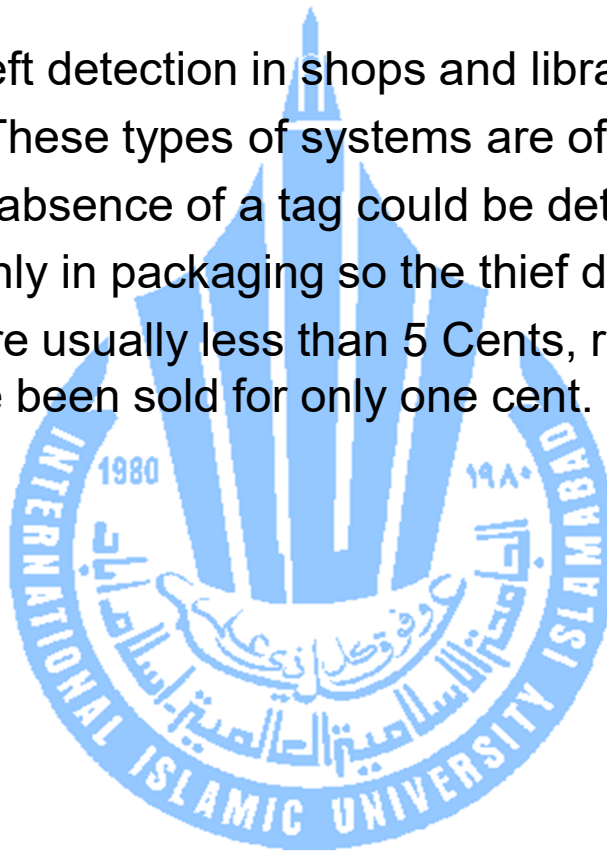
- Flat responsive electronic device.
- Typically no more than 0.3 millimeters thick.
- 04 Types of Smart Labels exist in market.

- **Electronic Article Surveillance (EAS)**
- **Smart labels for Brand Enhancement**
- **Smart labels for diagnostics**
- **Radio Frequency Identification (RFID)**



1) Electronic Article Surveillance (EAS)

- Primarily, Used for theft detection in shops and libraries.
- Do not contain data. These types of systems are often use '1-bit' tags
- Only the presence or absence of a tag could be detected
- Fitted at source - Mainly in packaging so the thief does not know they are there.
- Price of EAS labels are usually less than 5 Cents, recently some versions from China and Israel have been sold for only one cent.



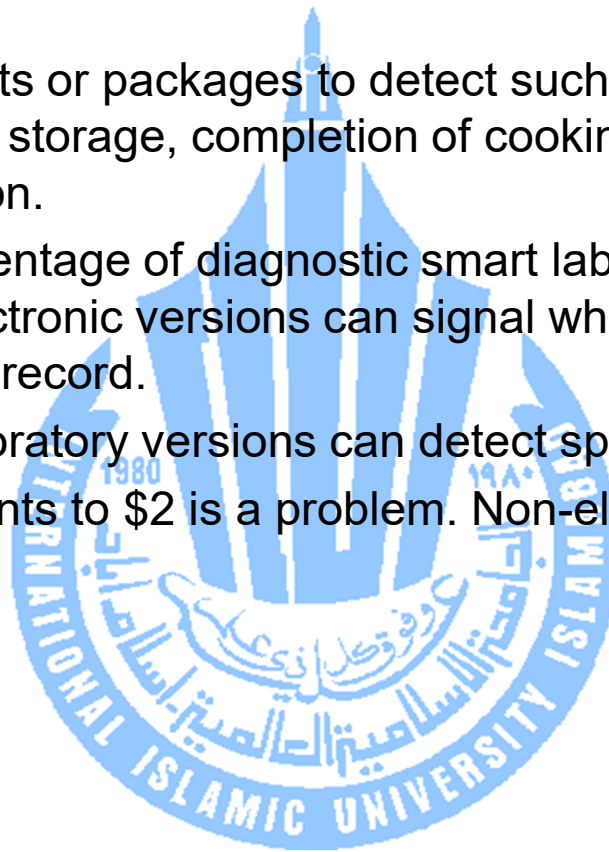
2) Smart Labels for Brand Enhancement

- In this context, Brand Enhancement is the adding of useful or attractive features to a product or its packaging via a label or insert,
- An example being disposable battery testers.



3) Smart Labels for Diagnostics

- Use of labels on products or packages to detect such events as dangerous overheating in transit or storage, completion of cooking, shock, tilt, vibration, completion of sterilization.
- Uses is very small percentage of diagnostic smart labels that are electronic is increasing because electronic versions can signal what is happening in real-time and most can store a digital record.
- For example, some laboratory versions can detect specific bacteria and viruses.
- However, price at 15 cents to \$2 is a problem. Non-electronic versions only cost a few cents.



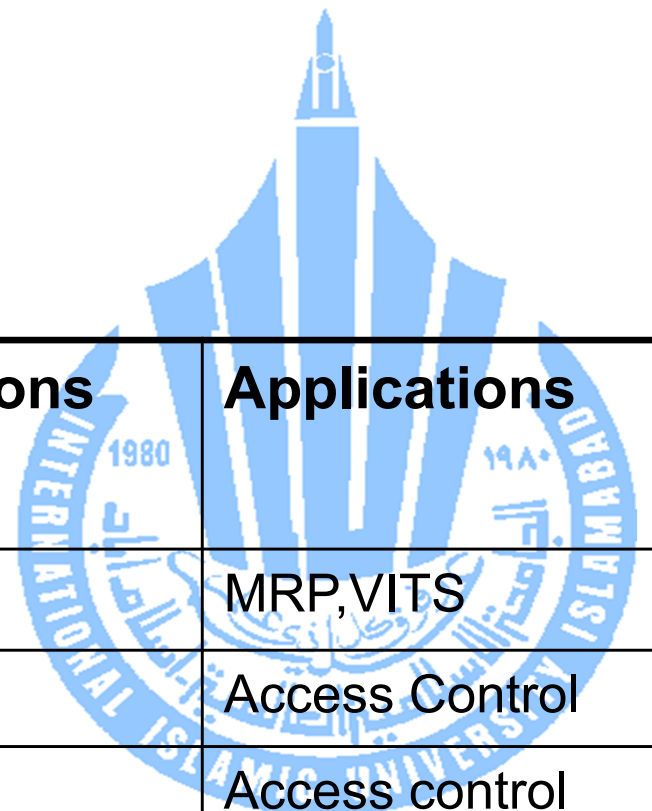
4) RFID

- **R**ADIO **F**REQUENCY **I**DENTIFICATION (**RFID**).



SCOPE OF PROJECT

- Primary developer of this technology is from (USA) 2nd World War and in Pakistan introduced by the end of 2003 and improving fastly.
- First Time in Pakistan used by NADRA
- In Pakistan there are only 03-05 Private/Govt departments are using RF technology
- New emerging long life RF Application are having very good market in Pakistan and all over the world



Organizations	Applications
NADRA	MRP,VITS
LMKR	Access Control
Mobilink	Access control
City Bank	Access control
NESCOM	In Progress
MIA Corporation	In Progress

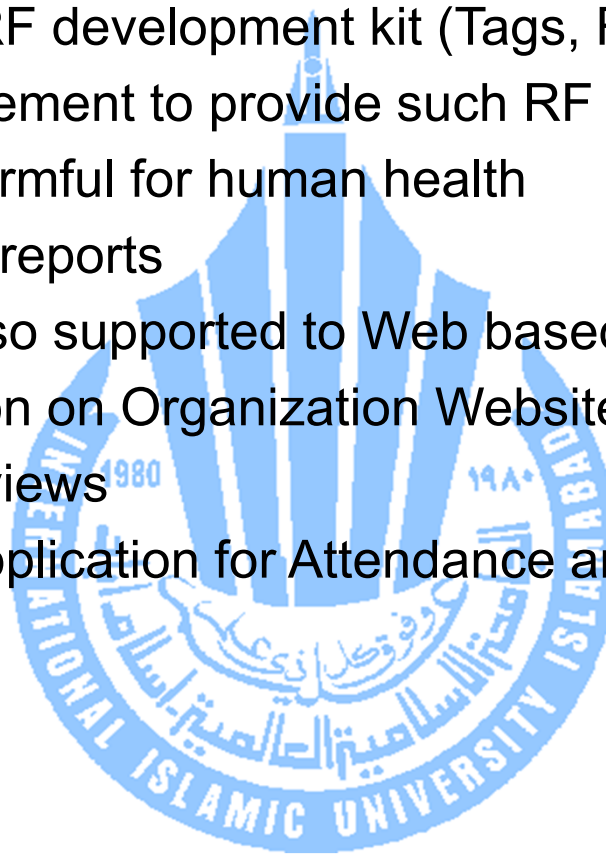
Existing System of MIA

- Organization has a healthy strength of staff Approx: 450
- Different departments manages their own Attendance
- Employee prove his/her availability by signing out on TWO Registers
- Some Staff get their pay manually & few by Banks (No Checks & Pay details)
- Late/Voluntarily sitting are being controlled manually for every employee
- Un-availability of Working hours details
- According to Organization building Internal structure it is difficult to find a person availability
- Manual + Obsolete and exhaustive (Standing in queue)
- General Manager /Regional Manger are not able to view (ONLINE) strength of their employees

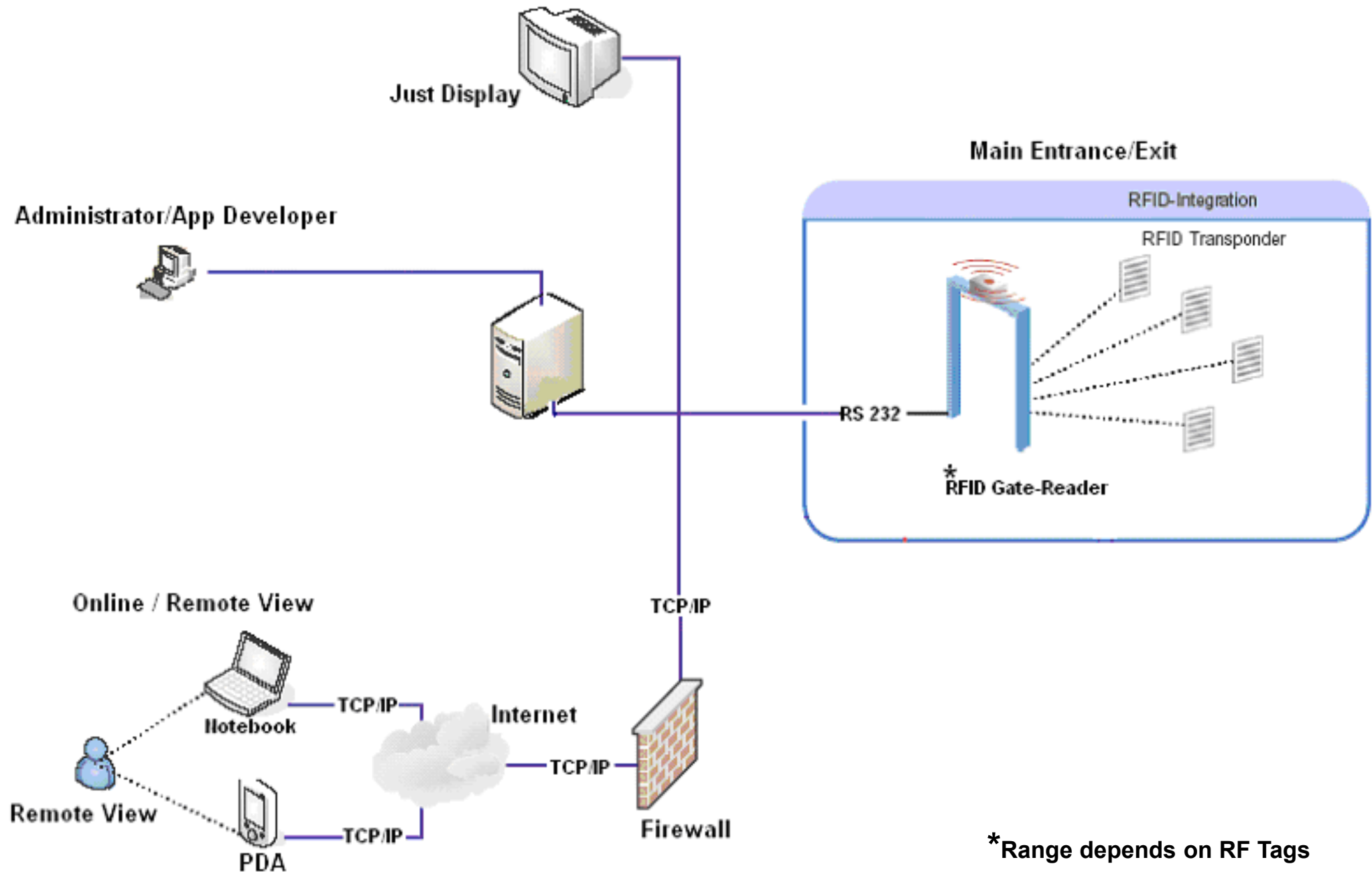
Proposed System for MIA

In order to design RF based Application

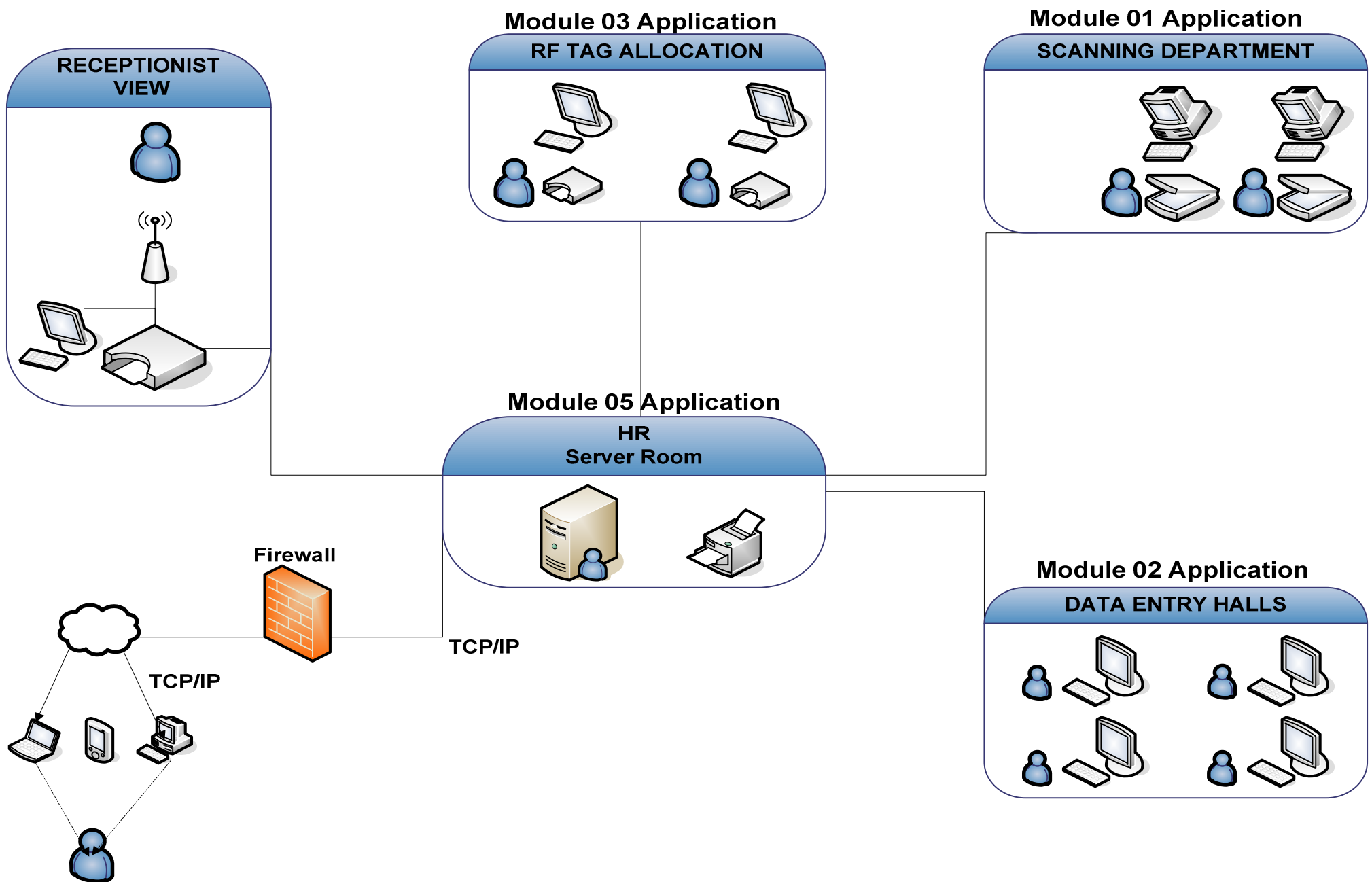
- We are bound to have RF development kit (Tags, Reader/Antenna, API's)
- MIA has signed an agreement to provide such RF kit
- Radio Waves are not harmful for human health
- Monthly, yearly detailed reports
- Complete Automated also supported to Web based views
- Employee facility to logon on Organization Website to check monthly details
- Support multiple dumb views
- Newly Automated RF Application for Attendance and Pay roll system



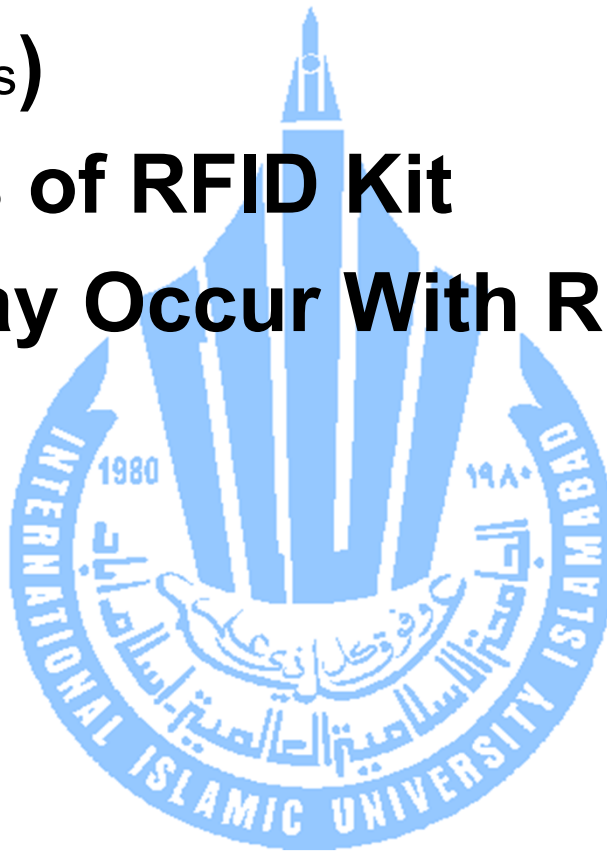
Proposed Solution



Proposed Solution in Modules

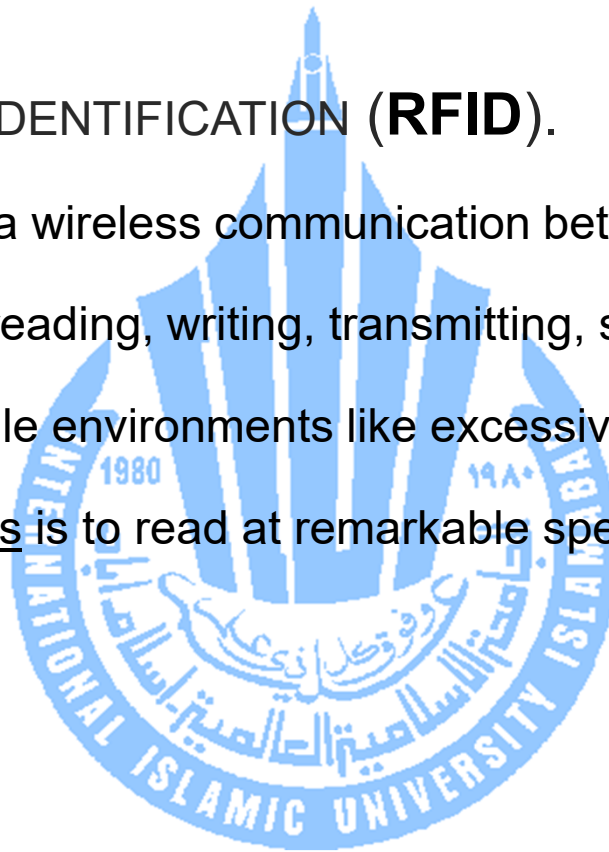


- **RFID** (Main Focus)
- **Components of RFID Kit**
- **Problems May Occur With RFID**



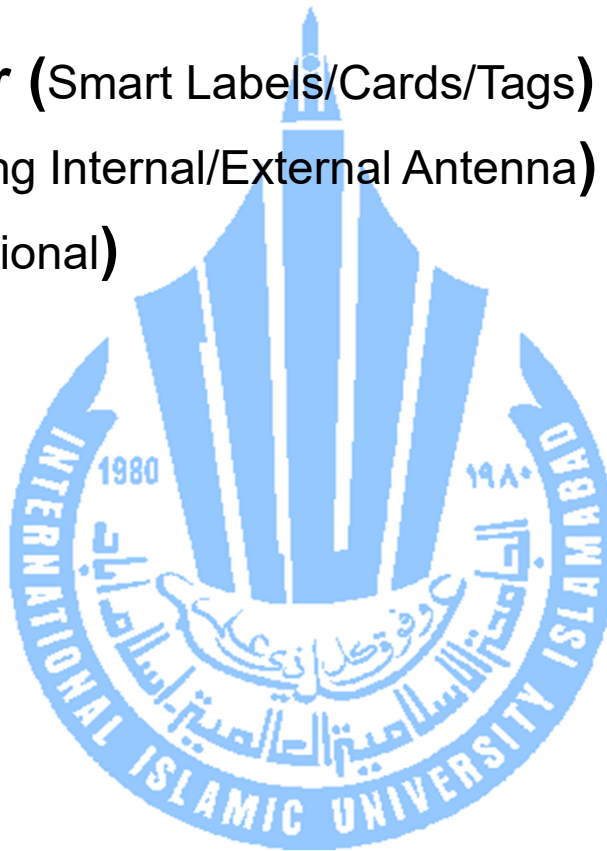
What is RFID?

- **R**ADIO **F**REQUENCY **I**DENTIFICATION (**RFID**).
- RF technology provides a wireless communication between objects (Tags) and readers.
- RF Tags are capable of reading, writing, transmitting, storing/updating info.
- Works effectively in hostile environments like excessive dirt, dust or poor visibility
- Most outstanding benefits is to read at remarkable speed i.e. Approx 1000 Tags per sec.



There are three main components of RFID kit

- **RFID Transponder** (Smart Labels/Cards/Tags)
- **RFID Reader** (Having Internal/External Antenna)
- **RFID Antenna** (Optional)



- Small silicon microchip attached to an antenna. Tags holds unique data and other unique attributes of the item.
- Size of Tag: Visiting card to Full Stop or depends on the size of the antenna.

There are Two (02) type of Tags.

1). Passive Tags

2). Active Tags

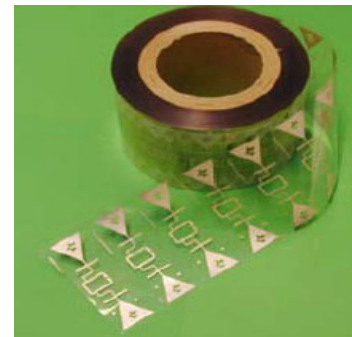
After merging Tags into products



RF Tags



RF Tag Rolls



Hitachi product Mini size of Tag

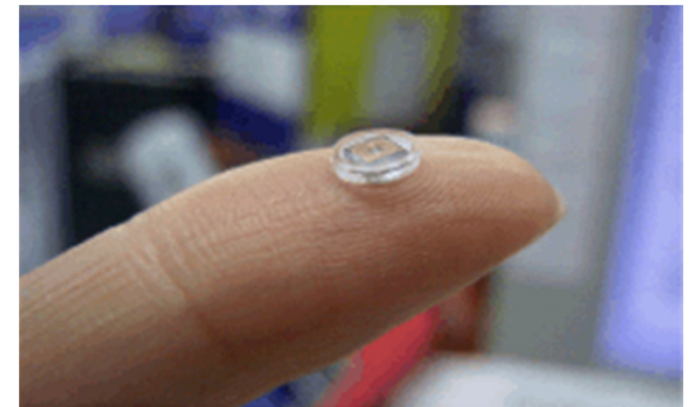


1). Passive Tags

- Tags which are powered solely by the RFID Antenna/reader.
- When the power to the silicon chip on the Tags meets the minimum voltage threshold it require to turn on, the silicon chip.
- Range is usually limited to several meters.
- The amount of data can be **up to 128 bytes** for Passive tags
- Tags are usually made to work at specific frequencies which are licensed free.

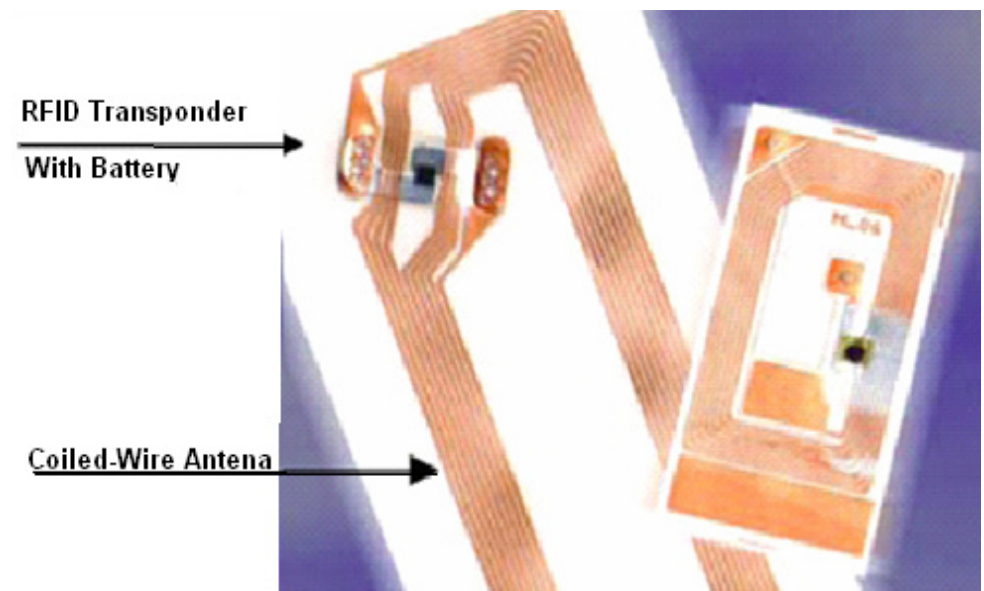
- Low Frequency (LF) 125-135 KHz
- High Frequency (HF) 13.56 MHz
- Ultra High Frequency (UHF) 868-930 MHz

- NADRA uses 902-928 MHz freq for VITS



2). Active Tags

- Having their own power source, so they can receive a weaker signal.
 - Its ranges of many tens of meters and even hundreds of meters.
 - The amount of data can be **up to 128 Kbytes** for Active tags
 - Its cost more because of their size and sophistication.
 - Battery life can also limit the life of the Tags.
 - Most common global frequency for active Tags - 433MHz
 - In special situation of very high-speed reading of Rail / Road vehicles (Non-stop tolling).
-
- Ultra High Frequency (UHF) 868-930 MHz
 - Microwave 2.45 GHz & 5.8 GHz



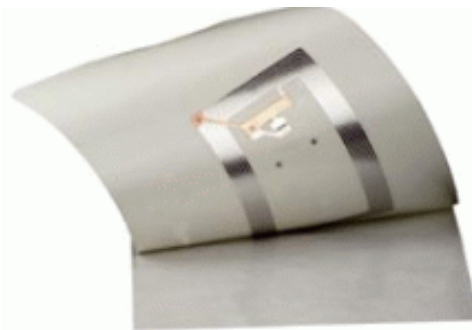
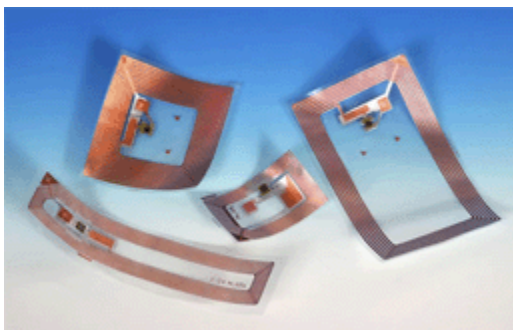
Tags with Chip Vs Without Chip

Tags with Chip

- Chip Tags have more cost and more data.
- They can hold larger amounts of data.
- Chip Tags are not usually available below 42 PKR (70 cents) if ordered in limited quantities.
- These uses where data handling is the primary challenge.

Tags without Chip

- They are much cheaper but less quantity of data.
- Chipless Tags are usually 20 PKR (30 to 40 cents) even for orders as few.
- Chipless Tags have the price advantages over chip Tags.

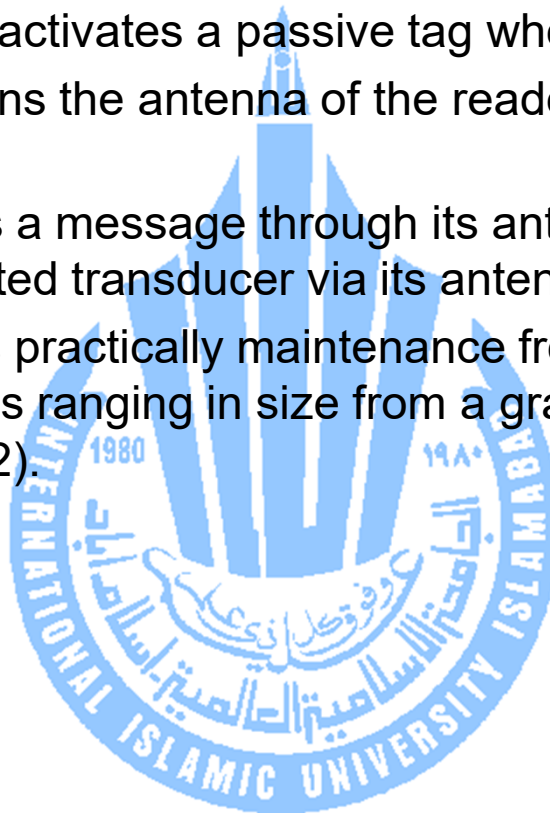


Tags Difference

	Active RFID	Passive RFID
Energy source	Internal to Tags	Energy transferred using RF from reader
Battery	Yes	No
Signal strength	Very low	Very high
Range	Up to 100m	Up to 3-5m, usually less
Multi-Tags reading	1000's of Tags recognized – up to 100mph	Few hundred within 3m of reader
Data Storage	Up to 128 Kb or read/ write & search	128 bytes of read/write

RFID Antenna

- Antenna attached to a reader to transmit an electromagnetic fields.
- Electromagnetic field that activates a passive tag when it is within reading range.
- During rewriting applications the antenna of the reader acts as a relay device in the reverse direction.
- The reader communicates a message through its antenna, which Transfers and stores the new data to the activated transducer via its antenna.
- The RFID tag's antenna is practically maintenance free and can be configured in a variety of shapes and sizes ranging in size from a grain of rice to the size of a brick (Zebra Technologies, 2002).



RFID Reader

- May be integrated into handheld device or may be stationary and positioned at strategic points.
- Handheld readers offer portability and stationary devices offer a larger reading range.
- Receives instructions and information from Antenna.
- The Antenna (Scanner's) information is then converted into a digital format by reader.
- And computer can then use for data Analysis, Recording, and Reporting.
- There are Readers today that can simultaneously read 100 to 2000 Tags Per Second.



IF5 Fixed "Smart" Reader



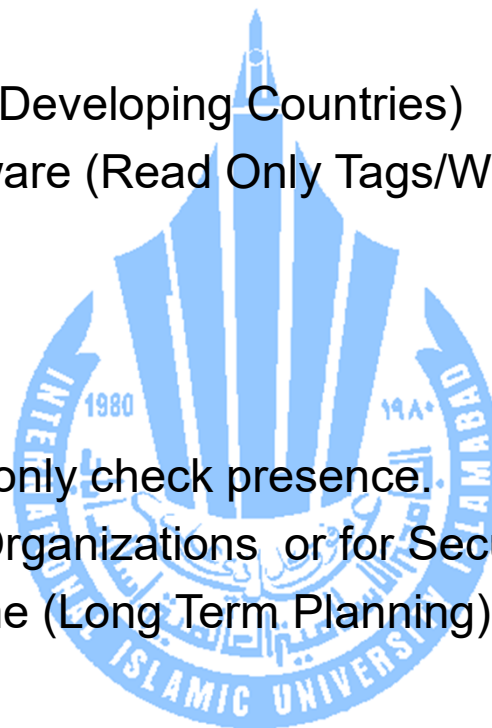
Intermec IP3 Portable Reader

Expected Problems:

- Tracking People's Freedom
- Expensive to Use (Only for Developing Countries)
- Life time of Hardware/Software (Read Only Tags/WORM Tags)

Solutions:

- Inform the Person, uses to only check presence.
- Ideal for Well Established Organizations or for Security / Military Purpose
- Good Decision at Good Time (Long Term Planning)



RFID VARIOUS APPLICATIONS



Active Tags Application (ATA)

(Depends upon Requirements and Cost)

Passive Tags Application (PTA).

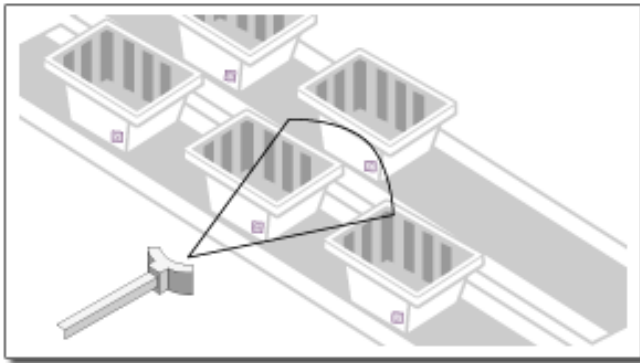
(Depends upon Requirements and Cost)

Where RFID can be implemented?

- Airlines and Airports
- Animals and Farming
- Financial and Security
- Healthcare
- Land and Sea Logistics
- Laundry
- Leisure
- Libraries and archiving
- Manufacturing
- Military
- Passenger Transport
- Retail/ Packing



RFID VARIOUS APPLICATIONS



Pharmaceuticals/ Food/Hotels
and package (PTA)



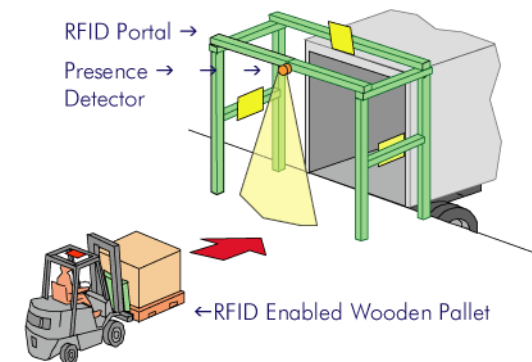
Special Children Schools
(ATA)



Attendance/Pay roll
(PTA)



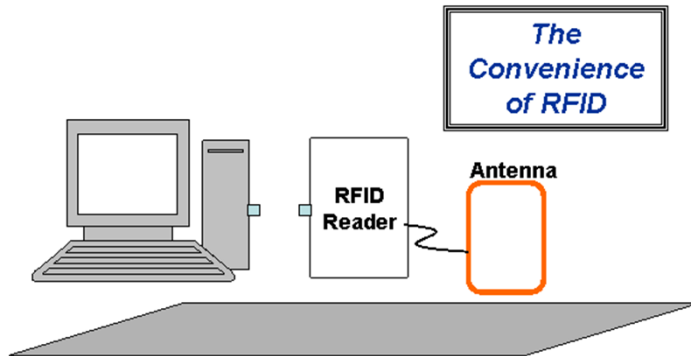
Special Children Monitoring
(ATA)



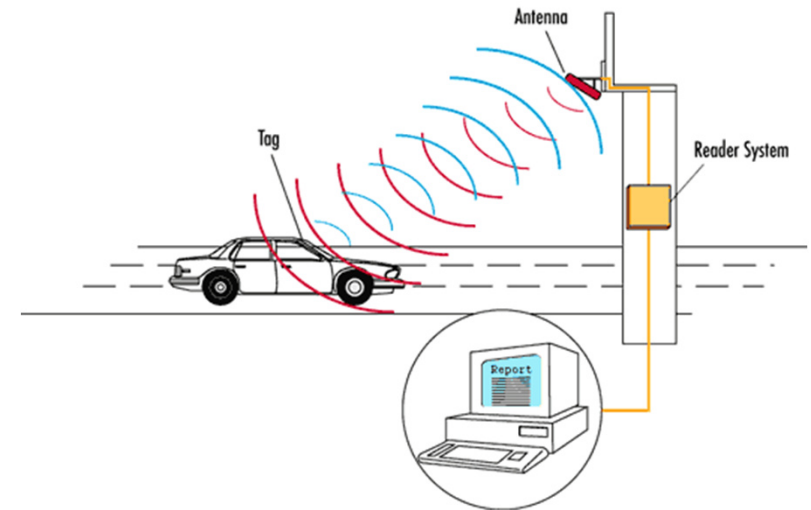
**RFID Conceptual Physical Design
for Warehouse Management**

Question & Answer

Baggage Identification System (PTA)



Car Identification System (ATA)



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Questions & Answers



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FINAL PROJECT PROPOSAL



ONLINE ATTENDANCE & PAYROLL SYSTEM FOR MIA ENTERPRISES THROUGH RFID (OA&PS RFID)

Presenter:

Muhammad Abid	Reg .No: 998/cs/mcs/05
Khurram Nazir	Reg .No: 987/cs/mcs/04

DEPARTMENT OF COMPUTER SCIENCES
FACULTY OF APPLIED SCIENCES
INTERNATIONAL ISLAMIC UNIVERSITY ISLAMABAD
2006

Preamble:

We assigned a group of 03 Lecturers and 01 Associate Professor; here are the name and discussion comments of the project.

1. Mr. Zulqarnain Hashmi (Lecturer)

The project is market base, May be assign.

2. Mr. M. Nadeem (Lecturer)

Discussed with the students, may be assign.

3. Mr. Sameer Akram (Lecturer)

Discussed with Only one student (Khurram Nazir), Idea is good. Project includes Attendance and Payroll Systems. But student did not give Payroll System Details. He needs to submit the detail of Payroll System. And Project should be actually implemented in MIA Corporation. After submitting Payroll System Details project may be assigned.

4. Mr. Shakeel Ahmed (Associate Professor)

Discussed with the students, Idea is not bad. But I have some concerns related to scope of this Project. Students are advised to talk with Dr. Afaq (HEAD Telecommunication & Engineering). If he (Dr. Afaq) recommends then Project may please be assign.

Dr. Afaq (Head of Telecommunication & Engineering)

We assigned to discuss the scope of Project by the “Head of Telecommunication & Engineering” Dr. Afaq.

“After discussing, we have come to reach this point that we will have encrypted information in a file and we will simulate the result for university purpose not for the organization. This will be the other security layer in our project, It increases the security because it will eliminate the chances to mark attendance of that person in MIA corporation either he has RFID based card of same frequency. And if employees of MIA Corporation visit to other organizations there attendance will never mark in their database because of the encrypted information.”

-----In the Name of ALLAHA the Most Beneficent and Merciful.-----

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1 Introduction

The present time demands from Universities and Researchers to Develop, to think and to work on according to most modern and demanded Technology. Especially in Computer Sciences fields, every day a new technology is coming. And every eve is going to off a technology. It has been identified that Developing Countries progress, their technology, and their approaches are far away from the modern world and modern technologies. In the 19's Century it was a quote that "Developing Countries are far away from Developed Countries from 80 to 100 years."

Now the time has been changed. World has become global village. Internet has been introduced, reducing gaps between the nations and expending very fastly. Internet provides new thoughts, techniques and technologies to all over the world. It provides equal opportunities to work and to think according to most modern ways. Now development of a country depends on its own nation or on its own interest.

The Proposed Project "Online Attendance & Payroll System for MIA Corporation through RFID (OA&PS RFID)" is based upon new and most demanded technology that is "Radio Frequency Identification (RFID)". That fulfills requirements of MIA Corporation (Islamabad Region) Attendance and Payroll System. MIA Corporation may enjoy a lot of advantages with this technology. One of them is MIA Corporation may enhance this technology by using on their own product, having front of them a model of this project.

This project uses RFID Kit and some other hard wares to successful implementation. So it has been confirmed and written statement has been got that MINIMUM HARDWARE cost will be fulfill by the MIA Corporation. After completion of Project Hardware would be property of MIA Corporation and software would be freely available ONLY for MIA Corporation. It also has been discussed with both students (Mr. Khurram Nazir and Mr. Muhammad Abid) have copy right reserved for this software. Hardware and general information about organization would be available as student's desire.

2 Introduction to organization

An Engineering base Organization "MIA Corporation" a Private Limited Company incorporated in 1992, under Companies Ordinance 1984 Pakistan. Organization has different branches in different cities of Pakistan and also in Middle East. Its corporate office exists in Islamabad.

MIA Corporation concerned with developing an applying different approach in the field of Engineering and Hardware disciplines. It provides the solution and fulfills the most demanded and valuable contemporary Engineering, mechanical problems.

An Engineering based organization provides total HVAC solutions for Residential, Commercial as well as Industrial Application. Having a healthy amount of skilled workers, brilliant team of professionals dedicated to Business Management, Finance and Engineering Services.

Scope of Services:

- Design and Project Management Services.
- Supply of HVAC Equipment.
- Supply of Refrigeration Equipment. (New Product).
- Installation, Testing & Commissioning.
- Warranty and after sales services.

According to Organization demands it has established separate warehouses in different cities of Pakistan. In order to ensure prompt deliveries of the product, Company have warehousing facility adjacent the regional offices.

Warehouses:

Islamabad (north region):

Industrial Area, I-9, Islamabad:	16000 Sq. ft
Humak Model Town, Islamabad:	18000 Sq. ft

Lahore (central region):

Shan Centre, Queen's Road, Lahore:	9000 Sq. ft
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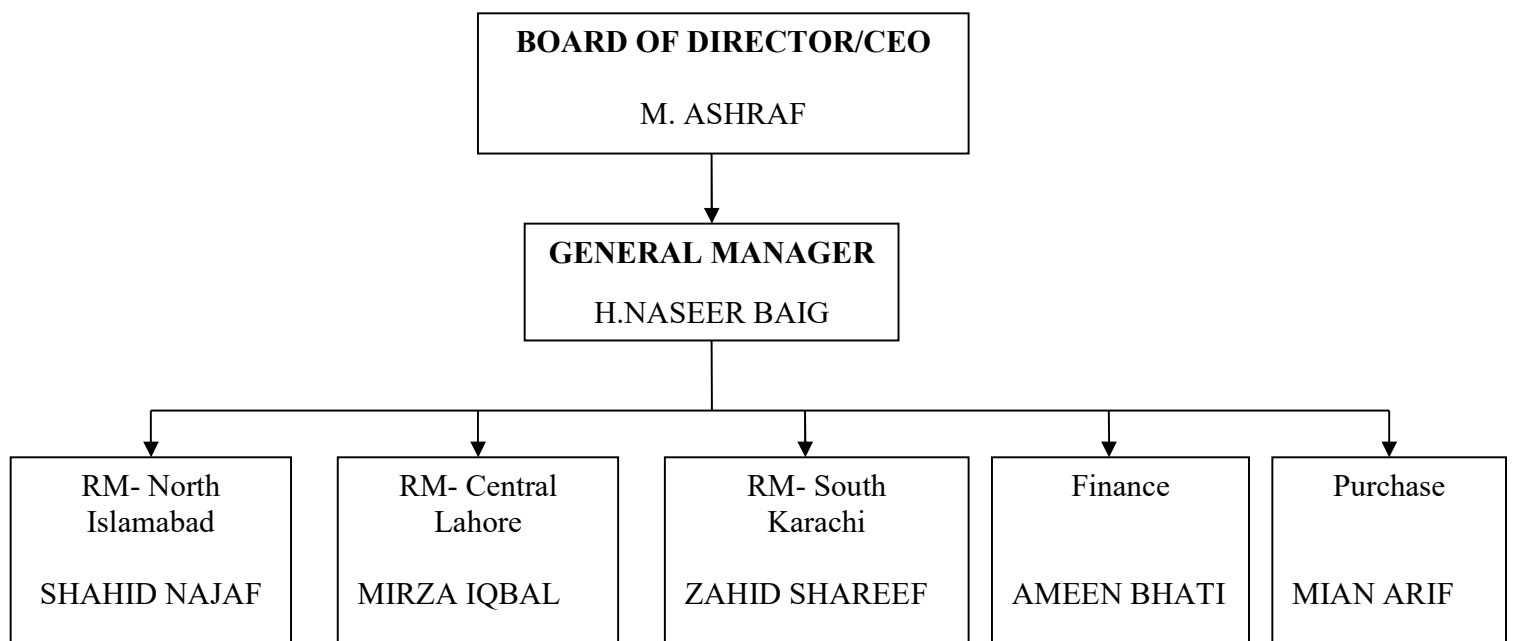
Karachi (south region):

Defense Phase-V, Karachi:	4500 Sq. ft
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2.1 Organization vision

- To be a dynamic and Leading HVAC Organization with Vision & Strengths of quality.
- To be an ultimate choice of the customer with reliable product at affordable cost.
- “A total customer satisfaction”, is pre-determined goal, focused by the management, to make it thus attainable, our SOP is implemented.
- In order to ensure full utility and comfort to the users, a team of qualified & factory trained Engineers are just one call away from the customer.

2.2 Organization Chart



MIA Corporation (PVT) LTD Organization Chart 2.1

2.3 Domain of Problem (North Region: Islamabad)

MIA Corporation concerned with developing an applying different approach in the field of Engineering and Hardware disciplines. It has 03 main regions. Approximate 03 branches of each region and having more than 500 employees in each region.

We are working for MIA Corporation Northern region Islamabad. Our domain of interest is Main North Region Islamabad not dealers/branches. MIA Corporation North region Islamabad has approximate 465 employees in different departments including (Mangers, Assistant to Mangers, Marketing survivors, customer support engineers, clerks, office boys, receptionists, telephone exchange, Departments (Hall) supervisors, clerks, data entry operators, drivers, sweepers and etc.

Organization provides its employees healthy competitive environments, suitable salary package, yearly leaves, Air line fare, house rent, late sitting with franchise benefits according to time.

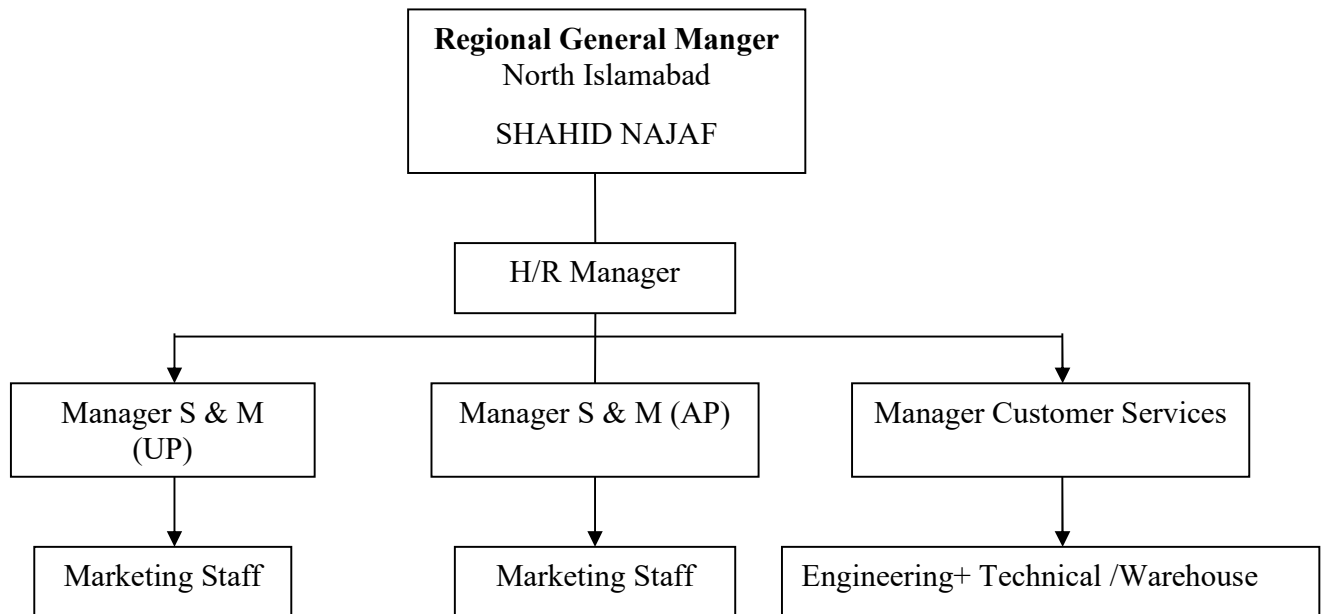
Organization is facing music to managing Attendance and Payroll system. Organization has it's a manual attendance and pay roll system. Some time Organization may wish to their employees to be available at sharp time or to have late sitting with late night. Normally office time considers Morning 08:00(AM) to 16:00(PM), and 05 working days in a week.

In a normal day two time attendance confirms by the all department's supervisors. Supervisors have attendance papers and submit their concerning (Hall/Departments) attendance to HR. In contemporary condition still Organization confuses about their employee availability.

And these tasks may be verifying by HR Administrators or concerning departments Managers.

MIA Corporation has its own website pending to be updated since last 09 months. Organization hopes for the best Automated Online Attendance and pay roll system. That may further extend for its other regional offices.

Regional Domain (North Region: Islamabad)



MIA Corporation (PVT) LTD Regional Domain Chart 2.2

2.4 Existing system

MIA Corporation has manually Attendance and payroll System. There is no proper mechanism to trace out an Employee's working time in hours+minutes+seconds. Currently deployed system is time consuming / obsolete / neither automated (Online) nor satisfactory. Managing more than 465 employee's attendance is a really bit rigid task.

2.5 Draw backs

- MIA Corporation has a healthy amount of employees that are divided into different Departments. Normally labors located in Warehouse department while other remains in Engineering, Technical, Managements and Finance departments. Managing Attendance in all departments is really a very critical and exhaustive job.
- MIA Corporation has its manual Employee Attendance System. An employee must have to sign out on a particular register for proving his availability in the Organization. There is no proper check and balance system (Coming and Out going time) for an employee.
- According to MIA Corporation Employees strength. Organization also has friendly atmosphere with different options for their employee to late sitting, voluntarily sitting and late with paid sitting. In Contemporary situation managing pays and attendance is being performed into to TWO Registers. One Register is reserved for morning attendance and other one is for after noon (after Lunch).
- There are different categories of employees in the organization few one are technical and others are in finance/management. Monthly salary is being paid by manuals system. There is a standard amount that is being paid by hand to its employee. And other one is getting paid by their banks after deduction of taxes, sur-charges and etc. Existing system does not provide ordinary checks/slips and monthly details.
- MIA Corporation has well name in the name of national and International Organizations. So Organization must meet with its Standard. Like Auto Attendance/Online product/Employee Tracing System.
- According to Organization internal departments a Staff may be available at different time on different departments or even out of Organization. Often seeker/visitor faces problems to meet with particular employees.

- Different departments have different kind of staff some one have Only Contractual and some one are Permanent. MIA Corporation has it's manually Employee Attendance System. According to Job status person are being paid. While contractual employees in/out time Organization demands critically. Still there is no proper check and balance system just having few registers.
- Existing system is exhaustive/time consuming, Not Online (Not Web supported) and not detailed.
- And finally obsolete/un-reliable (in sense of obsolete old system working with Registers & paper attendance...')

3 Proposed System

Organization was in hot water in respect to managing time +attendance and Payroll system. The Proposed system will be fully automated only for Islamabad region although it might be able to expand on other regional offices.

As we visit modern IT market we see different software developers have been developed different Attendance and Payroll systems. But the attraction of this Proposed system is to working with modern hard wares or technology.

To successful implementation of this project it is essential to have a RFID Hard wares (RF Kit). MIA Corporation now have been signed an agreement with us to provide lowest range (125-135-KHz Tags and Reader) hard wares.

Proposed system is most modern and highly demanded software in the market and required by this organization. Here are few steps those are spick and span to fulfill MIA Corporation desires according to managing Online Attendance and Payroll System.

As an employee applies for this Organization he is bound to fulfill Organization form. Now each employee form will be pass for data entry process. After passing from data entry phase, his provided information would be able to map with Unique (Read only) passive 125-135 KHz transponders or tags.

Such tagged card will be assigned to an employee for his Identification and as well for auto attendance. As employee visit from main entrance his info would be read from Its ID card. And Current date and time would be recorded in the database.

As employee visit from main entrance his info would be available at two different Dumb terminals (Main Entrance and in HR department).

Administrator or Application controller would be able to generate different kind of monthly salary/attendance reports with its username and password.

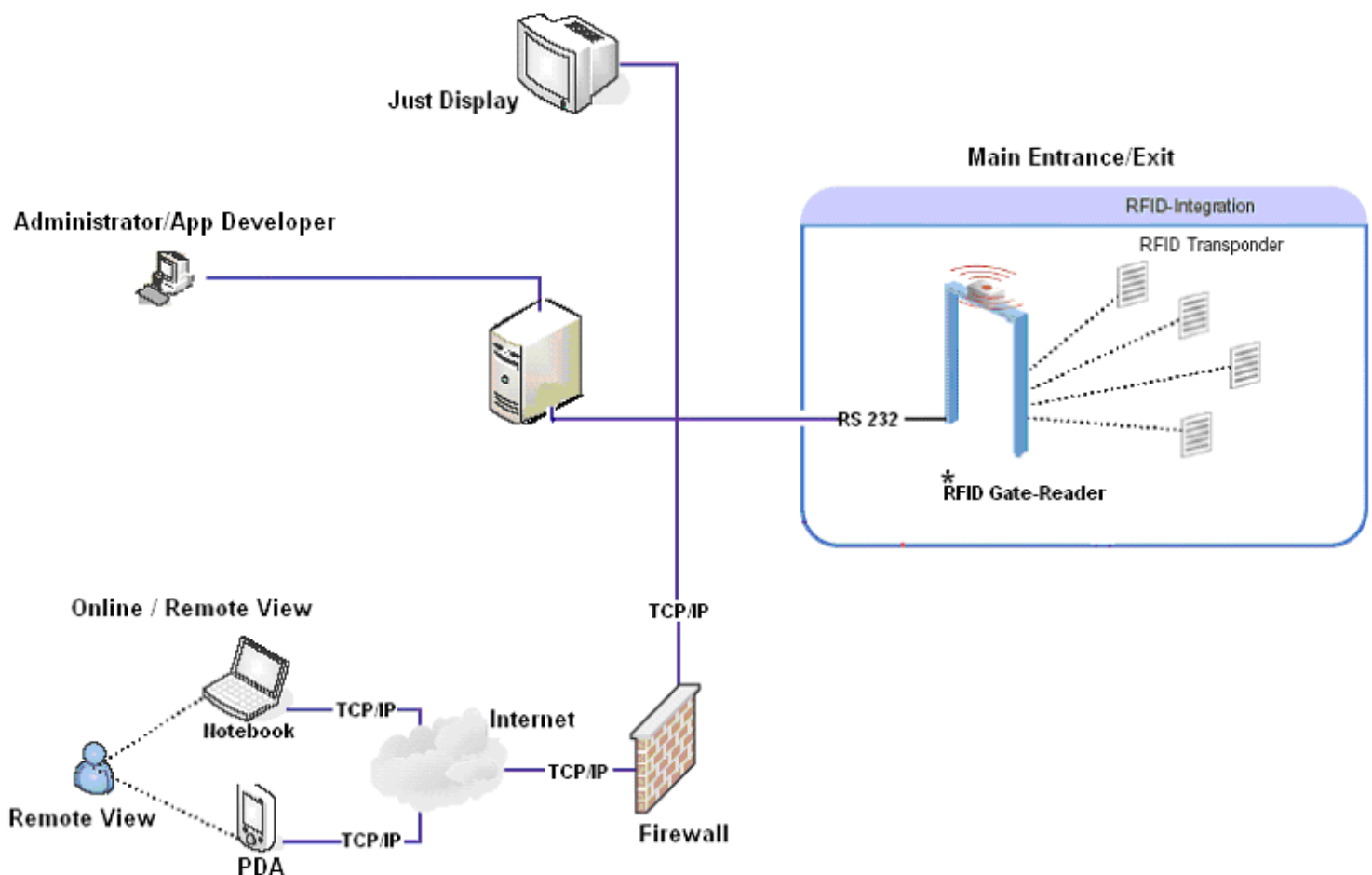
A web based application would be available every time on Organization website. So each employee would be able to check his salary/attendance sheet on the web.

3.1 Operational scenario

Module #01:

As a member wish or apply to join the organization its particulars taken by the Human Resource Departments. A new Employee fills a particular (Organization joining) form. These forms go to for data entry process (Scanning process) and finally employee info now available in the database for further process.

Each employee will have a unique identifier according to his department or skill. Unique Identification H/R Administrator may assign him an Organization based Identity Card. That card may have hidden Tag for managing his/her attendance.



*(Depends H/W Range)

Proposed system Diagram 3.1

Module #02:

All employees would have to keep the Organization Identity Card when they visit to Organization. When employee enters from main door (ONLY for Entrance), he/she would have/have not to show the card according to Tag/Reader Radio waves properties. Dumb devices (02 Monitors) would be in running position at main entrance and as well in H/R department. So (if required) Gatekeeper and as well H/R department would be able to identify the particular person have been joined the Organization.

Module #03:

In special case when an employee forgets/loose his Employee Card at home. Generally he would not be able to mark his attendance. If organization wishes to mark his attendance then she can turn ON the reader's/software option to operate in manual mode.

Module #04:

At any time or at the end of a month a user can check his attendance and pay details on the net. Further more payroll and attendance reports would be available for the employees.

3.2 Pay Roll System

In Online Attendance and Payroll System (OAPS) we manage Attendance by using RFID tags and Pays. In the payroll System MIA Corporation has two categories of employees.

1. Permanent employee.
2. Contract based employee.

[2.1 Long contract.](#)

[2.2 Daily wages.](#)

Computer Generated Checks

All type of Employees will have Computer generated monthly pay slips and also yearly reports.

Income Tax

Permanent type of employees and contract based employees (If he has contract more than 01 year) has annual salary less than 1,20,000 pay no tax according to the government of Pakistan taxations role.

And employees who have salary more than 1 20,000. They will have to pay tax monthly.

Overtime/Late sitting

Every Employee has option for late sitting or overtime. He will be paid according to his basic pay or MIA Corporation rule.

House Rent

According to MIA Corporation rule, offers house rents to Permanent employees according to their pay scales.

Medical Facility

Permanent employee and Contract based employees having long contract has medical facilities, to manage medical funds. These employees will have medical allowance.

Transport Facility

Permanent employee and Contract based employees having long contract are enjoying transport facility.

Bonus

Bonuses are being paid to both types of employees. Permanent employee will get bonus according to the basic pay, and Contract based employees will get according to their job nature.

Other Allowances

Further more allowances, MIA Corporation offer time to time different allowances and loan *Karz-e-Hasna* (For staff children marriages and other purpose) facilities to their Employees.

4 Software Features

In OA&PS RFID all the tasks are being completed in manual system will be able to perform in computerized fashion. This automated system has lots of benefits as compare to manual system. Some of these major benefits are,

- User friendly automated environment
- RFID Layers are not dangerous for Health
- Easy to Installation of hardware
- Abstract Data Security
- An Evolvable/Portable/Verifiable Standard Long life (up to 15-20 Years) Software
- A Completely Robust Software
- A Complete Reliable Auto Attendance Management System
- Every time Updated Information available
- A Web based login and reports viewing options for each/particular employee
- Ideal for International Conferences (Security & Guest Monitoring)
- A 100% Fault tolerance System
- Having Manual and Automatic Attendance options
- User defined pay cycles for time card entry
- User defined voluntary deductions and taxable status
- User defined deduction schedule and yearly progress
- Detailed employees history
- Computer generated reports & pay checks
- Less Electricity (Power)Consumption
- No more Signatures / files and 2nd time Attendance Registers
- A Standard/valuable/new technology meeting software
- No need to Show/Swap RF cards, No need more registers *_(Depends H/W Range)

5 Tools/ Implementation

Required Operating System

- Win 2000 NT/ Win XP
- RAM 128 MB (Min)
- Hard Disk 10GB (Min)

Required Software/Development Tools

- C# (Dot NET Framework)
- RFID Readers API's Supported to Dot Net Framework
- Oracle database
- Website
- Internet connection

Required Hardware

<u>Hardware Name</u>	<u>Quantity</u>
RFID kit (125-135 KHz Tags, Reader, Antenna)	01(Min)
Desktop PC P-III 2.0 GHZ	01
DUM terminals (Monitors)	02 (Guest Room / HR Dept)
DUM connector (06 Terminals can be connect with a CPU)	01
Scanner	01

6 Introducing to Sensor Based Technology (Smart Labels)

A smart label is a flat responsive device that is usually electronic. Most electronic devices are becoming flat because it is easier to lower their cost and fit them in awkward places such as packaging but we are not defining smart labels as generally as this. Smart labels are typically no more than 0.3 millimeters thick. We include smart tickets and laminates in the definition but not smart cards even if they work at a distance. These are a separate, but closely-related subject. Smart labels are becoming enormously important both socially and in terms of business potential. They can, for example, save lives and create billion dollar businesses.

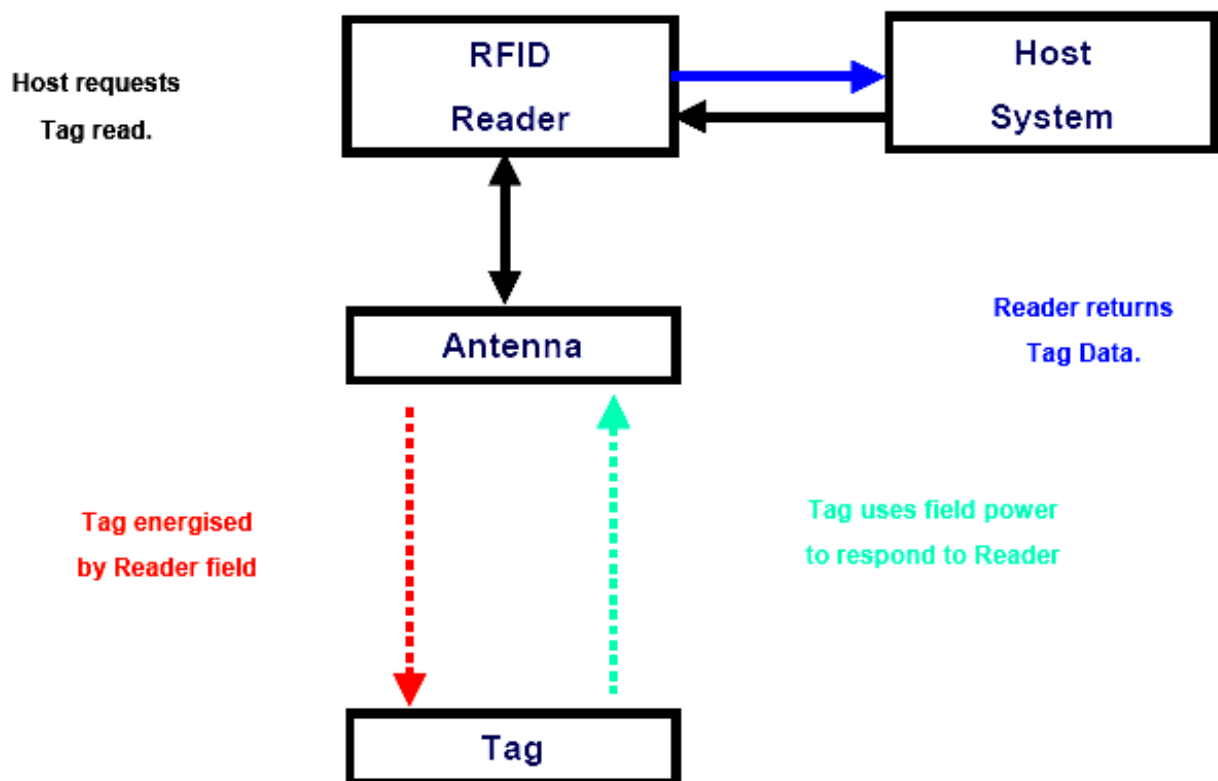
6.1 What is RFID?

Radio frequency identification (RFID) is automatic identification technology, the object of this technology is to carry data in an RFID card or tag (also known as transponder) data within the RFID tag may provide identification for an item in manufacturing, it is used to track variety of objects, including people or livestock, RFID technology relies on radio frequency between a card or tag and a reader in order to make identification. Since RFID does not require line-of-sight between the transponder and the reader, these systems overcome the limitations of other automatic identification approaches, such as bar coding. This means that RFID systems work effectively in hostile environments where excessive dirt, dust, moisture and/or poor visibility would normally interfere with the reading. One of the most outstanding benefits of RFID technology is its ability to read through these environments at remarkable speeds – responding in less than 100 milliseconds in most cases. Furthermore, RFID technology is completely automatic and transparent, eliminating the need to scan an object manually or activate a magnetic stripe, reader, or other contact ID technology.

The widest adopted classification is according to the underlying RFID-frequency. Limited by legal regulation and local integration capabilities, RFID-systems can be split up in Low Frequency- (LF), High frequency- (HF), Ultra-High- Frequency- Systems (UHF), and Microwaves. Each frequency bandwidth class has specific physical advantages and drawbacks, but HF and UHF are the most suitable classes for RFID-solutions.

6.2 How RFID works?

An RFID system consists of two major components – the reader and the transponder (also known as tags). The reader (or other antenna) performs several functions, one is to produce a low-level radio frequency magnetic field. When a tag is brought into the magnetic field produced by the reader the energy powers the internal circuit within the tag, this enables the transmission of the tag's memory content to the reader. The tag information is then received by an antenna within the reader; once the information has been processed the data is decoded and retransmitted to the host computer.



RFID Hardware communicating flow

Diagram 6.1

7 Components of RFID Kit

When comparing RFID-Systems to existing Auto-ID-Technologies in Pakistan (e.g. bar code), there are three major RFID components that effectively drive the success of RFID-solutions.

7.1 RFID Transponders (Tags)

RFID tags have two distinct classes:

7.1.1 Active Tag

Active tags refer to RFID tags which have their own power source, so they can receive a weaker signal from the interrogator (i.e. be further away), and the power source on the tag boosts the return signal. These types can have ranges of many tens of meters and even hundreds of meters, but cost more because of their size and sophistication. Battery life can also limit the life of the tag.

Key Features:

1. It can be read at distances of one hundred feet or more, greatly improving the utility of the device.
2. It may have other sensors that can use electricity for power.
3. It has Larger Data Storage Capacity.
4. Because of the large read range, Active Wave RFID tags can send data at pre-designated intervals or at certain locations.

7.1.2 Passive Tag

Passive tags refer to RFID tags which are powered solely by the RFID interrogator (Antenna/reader). The interrogator emits a radio frequency (RF), which powers the silicon chip on the tag when it is within range of the RF field. When the power to the silicon chip on the tag meets the minimum voltage threshold it requires to turn on, the silicon chip can then send back information on the same RF wave. Range is usually limited to several meters.

Key Features of Passive Tags.

1. The tag functions without a battery; these tags have a useful life of twenty years or more.
2. The tag is typically much less expensive to manufacture.
3. The tag is much smaller (some tags are the size of a grain of rice).
4. These tags have almost unlimited applications in consumer goods and other areas.
5. The tags have a life that usually outlasts the object that it is identifying.

Passive Tag Types:

1. Glass Tag
2. Tear Shaped
3. Clear Disc
4. LOGI Tag
5. World Tag
6. Volcano Tag
7. Wrist Band

Read only, Read/Write and WORM tags

- **Read only** tags contain a unique license plate number which cannot be changed
- **WORM** Write Once Read Many - enables users to encode tags at the first instance of use, and then the code becomes locked and cannot be changed.
- **Read/write** allows for updated or new information to be written to the tag.

Comparisons

Read only tags can be the cheapest, because they often require the least amount of memory, but they rely on an infrastructure and readily available database to retrieve useful information. Where this is not possible, read/write tags, which are more expensive (but do not need large infrastructure) and often used. In many cases these functionalities are combined with segmented memories.

7.1.3 Frequencies for Passive RFID Tags

The most popular frequencies used for passive (i.e. low-cost, modest functionality) RFID devices are 125-134 KHz (Low Frequency - LF), 13.56 MHz (High Frequency - HF), UHF (868 to 928 MHz - Ultra-High Frequency) and 2.45 GHz (Microwave). A popular frequency for lowest-cost RFID tags at present is 13.56 MHz. It is the basis for several international standards, economical electronics is available and it is a good compromise of tag size and range and the extremes of flooding a small volume with radio waves (LF) or transmitting thin beams (Microwave). Since fewer turns of antenna coil are usually needed, these can be printed (say PCB etched by the method used for printed circuit boards) when 125 KHz tags usually need copper wire. 13.56 MHz is already the agreed frequency for smart cards and one of the frequencies specified for airline baggage tags. It is popular for supply chains in Japan and at least one meter range can legally be achieved in most countries.

However, tags at UHF (around 900 MHz) and 2.45 GHz are being developed for general RFID uses. Many are already in use tracking vehicles, freight etc, particularly in the US (mainly UHF) and Japan (2.45 GHz only). For UHF, that often means 10 meters range and for 2.45 GHz several meters. Thus UHF improves range and gives a smaller tag than HF. Microwaves gives intermediate range but gives the smallest tag of all. The cost of the complete device depends on the size - the price of the chip alone does not change with frequency, but the size of the antenna, and hence cost of the complete tag and its outer protection does.

Any Standard for Tags?

Yes, the ISO/IEC has a standard for 126-135 KHz RFID tags and reader electronics. 15693 is a global standard for 13.56 MHz RFID tags and reader electronics. The ISO/IE C 15693 protocol is designed to conform to FCC (USA), ETSI (Europe) and MPT (Japan) regulations worldwide, allowing tags to travel around the world under optimal conditions for operation with region specific RFID readers.

7.2 RFID Antenna

The function of the antenna attached to a reader is to transmit an electromagnetic field that activates a passive tag when it is within reading range. Once a passive tag is activated it can transmit information from its antenna to that of the reader where it is processed. During rewriting applications the antenna of the reader acts as a relay device in the reverse direction, the reader communicates a message through its antenna, which Transfers and stores the new data to the activated transducer via its antenna. The RFID tag's antenna is practically maintenance free and can be configured in a variety of shapes and sizes ranging in size from a grain of rice to the size of a brick (Zebra Technologies, 2002). However, it is very common to see transponders and tag antennas packaged as smart labels (sealed RFID tags) consisting of an integrated circuit (IC) attached to an Antenna in the shape of a coil of wires as in Diagram-1

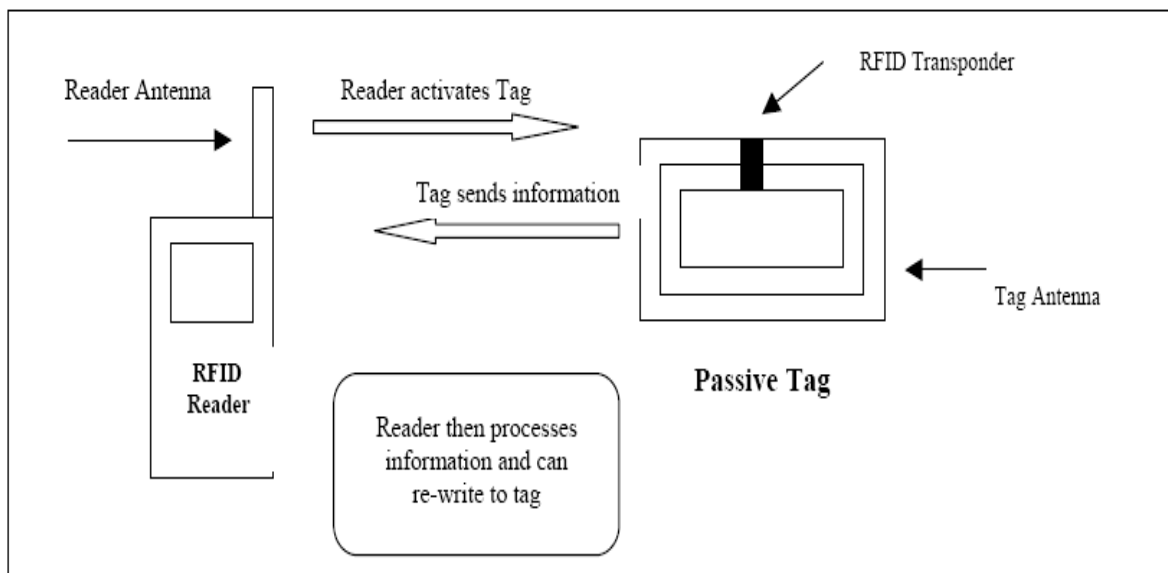


Diagram 1. RFID Reader and Tag Communication Flow Diagram

7.3 RFID Reader

Readers may be integrated into handheld computers or they may be stationary and Positioned at strategic points, such as a facility entrance or on an assembly line. The handheld readers offer portability, however, the stationary devices offer a larger reading range. As stated above, readers have an antenna for sending and receiving signals and a processor for decoding them. The reader receives instructions and information from the antenna through the scanner, which is a part of the reader that examines analog output from the antenna. The scanner's information is then converted into a digital format by the reader, which the computer or processor can then use for data analysis, recording, and reporting (CII, 2001). There are readers today that can simultaneously read 100 to 2000 tags per second.

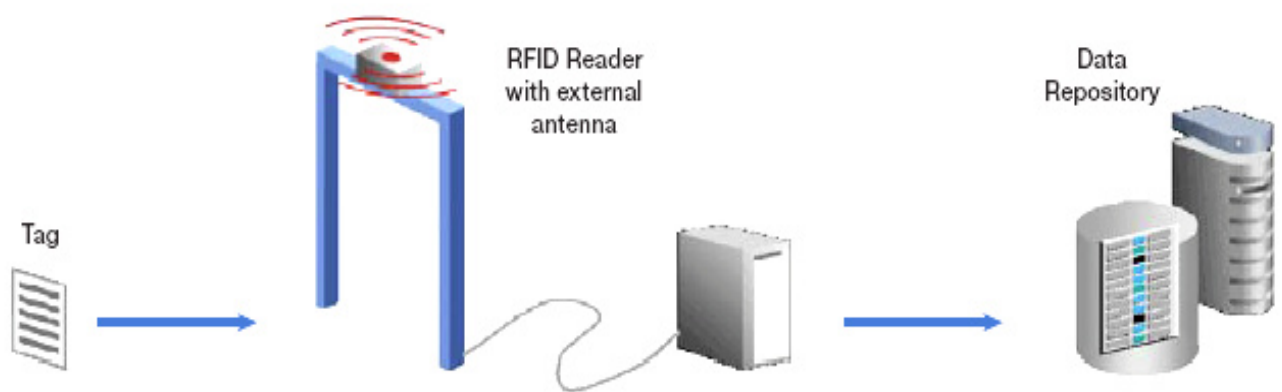


Diagram 02: RFID Kit in Progress.

What drives up reader cost?

In fact, there are four aspects that drive up the cost of interrogators for The Internet of Things:

1. Multi-frequency, multi-protocol.
2. Radio or hard-wired connection to the network in order to give real time at least some of the time.
3. Anti-collision (reading many tags in the field at one time).
4. The unprecedented and unmanageable volume of data generated, much of which is not needed, means the interrogators must process information before passing it on in simpler form.

7.5 RFID Transponders Required Power

The passive tag requires a minimum of -10dBm (100 μ W) just to power up and receives only 7dB more than the minimum requirement. This 7 dB margin is not adequate for real world situations.

RF noise or RF adsorption or reflection easily exceeds this margin. Experiments have shown that around 25dB is needed to penetrate and read tags in a case of water.

This illustrates why passive tags are so hard to read on certain products.

System Component	Power	Units
Reader RF output power	+30	dBm
Reader Antenna gain	6	dB
Path loss at 3 m	-41	dB
Tag Antenna gain	2	dBi
Received power at tag (sum of all above)	-3	dBm
Power to activate tag	-10	dBm
Margin in forward link	7	dB

RFID forward link budget Table 7.2

7.6 Benefits

There are many benefits to using RFID tags rather than the standard bar code. Bar Codes must be separated and scanned individually by a reader, whereas radio frequency tags can be scanned simultaneously all at one instant. Bar codes do not compare to RFID tags in terms of durability. Bar codes are usually displayed on paper products or hard metals that make them vulnerable to harsh environments. Bar codes are deemed dysfunctional when they have been torn, disfigured, or detached from their objects. Furthermore, RFID tags are read more accurately, 99.9% read rate compared to 80% read rate for bar codes, because they eliminate manual data entry and can be read through most all materials (Conway, 2001). 9 Bar codes have a storage capacity of approximately 20 data characters, while a Typical RFID tag can easily store between 8 and 32 MB of data. Furthermore, within the next five or six years it is projected that there will be a shortage of bar code combinations and products will either have to start duplicating code identification numbers or start adding an additional digit to their character strands. It may sound simple to just manufacturer bar codes with a longer identification number.

But when you add in the costs of upgrading readers and processors with software to handle the new character strands it makes more sense, and cents, to think into the future. RFID technologies are more reliable than bar codes because they have no moving parts and require no maintenance. RFID also provides higher security because each tag is extremely unique and almost impossible to duplicate (Durfee, 2003). Since RFID does not require a line-of-sight and has a greater read range than bar codes they prevent thievery and counterfeiting of products and goods. The continuous communication and rewritable capabilities of RFID enable electronic tags to be tracked in real-time through the use of GPS, and they can still be reused for other applications.

7.7 RFID Drawbacks

- Relatively high cost of tags
- Read/write range can limit application
- Multiple read/writes are time dependent
- Standards are still emerging and not mature
- Non-reads must be catered for within the system

8 Cost & Expenditures

RFID technology is rare in developing countries so its price may differ with other countries. Currently Active Tags and High range passive Tags are not available in Pakistan. Recently few dealers got their dealership with ZEBRA TAGS (USA). And they have 125-136 MHz and 13.56 MHz frequency Hardware.

As we mention above that Tags prices differ with technologies (Size, range, Memory and active/passive). Normally Active tags are not available less than 12 \$ (US Dollars). And a passive tag with chip (13.56 MHz) cost goes up to (03-05 \$ US Dollars, 180 to 300 Pak Rupees).

Same as Reader cost May very with tag's frequency range. Cheapest reader for (125-136 KHz tags) starts from 200 \$ to 100,000 \$ and some time more expensive. Antenna cost some time included in Reader and some time separate antenna require depends on application.

As concern with our project, MIA Corporation has approved to fulfill hardware requirements for 125-136 KHz READ ONLY Passive Tags application.

Case Study

In Pakistan NADRA is using UHF Passive Tag with chip (05\$ US Dollars, 500 Pak rupees) Application with YAGI ANTENNA (100-500\$ US Dollars) and Reader (5000\$ US Dollars, 300,000 Pak Rupees) for her Vehicles Identification & Tracking System (VITS).

9 Conclusion

The Project (OA&PS) is going to show its initial implementation. We are going to use lowest range of hard wares. In our project Tags cost may go 1-20 Cents (15 Pak Rupees) because of chip less tags and Reader cost may go 200-280\$ US Dollars (12000-18000 Pak Rupees). That is completely feasible to MIA Corporation.

OA&PS is initially would be first version of software for this lowest range of Hard wares. MIA Corporation has declared that in case of successful implementation they may be able to expand this project for UHF range hard wares.

RFID technology is not a replacement for bar codes. It is available in a mature and usable format with standards beginning to emerge. Unlike bar codes, RFID does not require line of site. Due to tag cost it is currently more applicable to the transparent tracking of reusable assets which have been previously associated with the products they hold. Successful project implementation is dependent on the quality of the systems integrator who must have goods business and process knowledge, technical understanding of the component products and wide integration resource capability.

Appendix-A

Latest Progress and Opportunities in RFID

While the potential for RFID applications appears virtually limitless there are only a few applications that have been implemented and are continually being used. In the year 2004 the most common uses for radio frequency identification technologies included: Transportation, materials management, and especially for security and Military Equipments delivery and **Access control**. Currently, there are a variety of other applications for RFID in agriculture, construction, and athletics.

A-1 Security

Many businesses use RFID to control access to hotels and business facilities by attaching a tag to an employee's room card or ID badge. This application is also becoming more common in nursing homes and hospitals where the management and tracking of individuals is very important, and alarms are more discrete. Other security features include RFID chips embedded into automobile keys that enable the car only to start if the key has the proper chip embedded into it. Video stores and libraries are also applying radio frequency devices to checkout rentals and detect stolen or misplaced items. Law enforcement officers are now able to track credit cards, jewelry, vehicles and artwork by radio frequency tags embedded in these objects.

A-2 Sports

Golf ball identification and tracking devices have been implemented for accurate Identification purposes; however, there may be a better market for the tracking of baseballs in today's homerun hitting bonanza. Another use for RFID tags in athletics is implemented in almost all major track and field events. "Road races," running races or marathons in the streets, use shoelace RFID tags to get race results of runners as they cross the start and finish lines where there time is officially kept regardless of when the runner begins the race. These technologies are also being used to track athletes to verify that the path traveled is the same as the course defined by the race officials.

A-3 Documents Controlling

Different Military, Government Agencies, Research Based Institutes are those organization their work highly depends upon the papers. They have different tasks like transmitting files, Display drawings, Arms Simulation plans and other specifications, jobs change orders and billing even now can be controlled and their record can be maintain with this Highly demanded technology.

According to security and management when the cost of RFID technology is more Reasonable, applications that embed tags into construction documents, files.

Or file folders would significantly reduce the amount of time and money spent managing files, and until the industry is paperless there will always be a way to more efficiently manage documents.

Tiny integrated circuits (ICs) can be made into labels or embedded into the paper itself. ONLY 0.4-mm square and is wireless accessible at 2.45 GHz allowing files to be secured and managed more efficiently. A document equipped with RFID tags enable files to encode file identification numbers making them easier to locate and secures them from being removed without authorization from designated filing areas. This technology could be implemented with automated checkout capabilities with RFID tagged employees or an alarming system that identifies documents leaving a secured area.

Current Risks and challenges include the movement towards electronic file transfers and small reading ranges of the technology. The Hitachi chip is stated to have a read range of one-foot, indicating that readers would have to be installed in each file cabinet drawer and within close proximity to entrances to implement such technologies. Very easy to install and control.

A-4 Global Positioning System (GPS)

Global Positioning Systems (GPS) have revolutionized the means to accurately Locate and identify objects on the earth's surface using a system of satellites in space and transmitters and receivers on earth. The combination of GPS and RFID identification tags has made "real-time" tracking a reality. Materials and assets can be identified and tracked as they are installed or transported.

A-5 Airports – Security and Baggage Handling

RFID tags could be used to track and identify airline luggage and passengers Increasing national security, speeding up luggage sorting and transfer, and decreasing expenditures resulting from heightened security measures. The International Air Transport Association (IATA) believes this technology has countless potential benefits for simplifying passenger travel for airports and airlines (Conway, 2001).

The major advantages of RF technology in baggage handling are an increased Journey speed of luggage as a result of faster read rates and elimination of human Intervention in misdirected bags and security procedures. Airline travel consists of 100 million customers each year making the cost of the tags and the read range of baggage a great concern. However, if the tag's cost can be reduced from its current price around 50 cents down to 15 cents and the IATA makes the 2.45 GHz frequency standard (doubling the read range of 13.56 MHz frequencies), this tagging application in the airline industry would takeoff.

In November 2001 Jacksonville International Airport became the first airport to begin constructing and implementing the world's first all-radio frequency baggage tracking and identification system (IIE Solutions, 2002).

The RFID tagging system will make curbside check-in and security inspections more efficient by allowing readers to identify not only where the baggage is going, but also passenger information and a detailed history of the bag. However, until the entire airline industry follows suite there is going to be a double standard for those implementing new technologies.

A-6 Future of RFID Applications in Pakistan

According to our country, Pakistan's Geographically, Economically and defense point of view it demands highly RFID Applications. Almost all daily life's fields as mention above, there are a high demand on standardization, economy, and versatility of radio frequency identification tags. Future ideas are derived from many of the new and current uses of RFID. However, there are several ideas that Islamic Republic of Pakistan even other government and non Govt organizations may implement in different areas of interest.

A-6

Different Hospitals, Medical Institutes **Especially HOLY FMILY HOSPITAL Rawalpindi** desires and waiting such type of Technology. Medical applications include allowing restricted access and tracking patient's and guest with authorized wristbands through hospitals.

A-7

The supply-chain and sales businesses have adopted RFID to help persons such as pharmaceutical sales representatives to track inventory by simply walking into the drug storage room. The sales representative could determine which products are moving and assist drug companies how to better market their products.

A-8

Government agencies could identify and track any suspected Personality. RFID applications are highly demanded in Pakistan Research and Defense based Institute (PIEAS, PAEC, HMC, KRL, IIUI and etc).

A-9

RFID also provide help to manage prisoners in Jail and to have (keep) tight security surveillance on convicted criminals.

A-10

The automobile industry may have a jump start in tracking vehicle's maintenance and service records by means of smart tags located within the vehicle's tires. Such information in conjunction with the car's odometer could track the mileage of the vehicle and alert the driver when the tires need to be rotated and the engine oil needs changing.

A-11

The Pakistan military should Implement this techniques to identify trained dog tags to easily identify troops and to track other important information such as medical history.

A-12

RFID technologies upgraded to produce instant location information, Real Time Location Systems (RTLS), may influence food distributors and grocery store chains to be completely RFID oriented. RTLS allows a reader unit to "see" the actual location of an RFID tagged item, without the tagged item being near the reader. Using special readers placed around a property, tags are located using a triangulation system.

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